

Diversity-Aware Aggregation in Random Forests

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1. Abstract

Random forests are a widely used ensemble learning method. However, many trees within a random forest become highly correlated, especially when trained on similar samples or when the same subset of features are close to the root. These correlations lead to redundant votes and limit the potential gains from ensembling. In this work, we propose a diversity-aware random forest that enhances ensemble learning by modifying the aggregation stage rather than the building process. Our method utilizes a structural similarity metric and a prediction agreement metric. Based on these measures, each tree is assigned a diversity score reflecting its redundancy. Utilizing these diversity scores, we replaced uniform majority voting with weighted voting in which trees with higher diversity contribute more to the final prediction. This resulted in two random forests, and these weighted approaches were compared against standard random forests, utilizing the Student Performance dataset from Kaggle.

2. Introduction

Random forests are a popular ensemble learning method due to their overall strong performance, robustness to overfitting, and ease of use across a range of tasks. By aggregating predictions from many decision trees on samples with replacement and random feature subsets, random forests aim to reduce variance and improve generalization compared to individual decision trees. However, standard random forests provide limitations caused from the high correlation between the trees. This issue arises when similar training samples are selected during a bootstrap sample or when a subset of features repeatedly appears near the root of several trees.

High correlation among trees reduces the effectiveness of ensembling due to redundant trees contributing little new information to the final prediction. In these cases, uniform majority voting treats all trees equally, regardless of whether a tree provides unique predictive behavior or a repetition of patterns already recognized by other trees. Improving how diversity is leveraged within random forests is an important problem, as it can lead to better generalization performance without increasing model complexity or training cost.

Therefore, we propose two random forest variants that are diversity-aware when aggregating predictions, meaning that they favor more unique trees, whether measured by prediction agreement or structural similarity. The input to our algorithm is an instance of a student. We then use our random forest algorithms to output the student's predicted performance in class: high, medium, or low.

3. Related work

Several papers have targeted the issue of generalization in random forests. Liu et al. explored the idea of increasing tree diversity in order to improve generalization through building complete random trees, finding that randomization methods such as bootstrap sampling may impair tree growth and limit tree diversity. Similarly, Zhang et al. introduced a subsampling technique that assists in implementing an initial generation of diverse datasets for training. To increase model diversity, they introduce an ensemble based optimization and an anomaly pruning strategy. Abellan et al. explored imprecise probabilities, as a new split criterion to increase diversity during the training process. By representing probabilities as intervals rather than precise values, their method increases uncertainty at split decisions, leading to more diverse tree structures. Ko et al. proposed a Diversity Index (DI) to characterize the diversity of data quality

(Qi) among selected training samples required for constructing a classification model of tree genera. Karabadji et al. introduced a genetic algorithm based optimization approach to balance accuracy and diversity in random forests. Their method simultaneously aims to improve individual tree accuracy, maximize diversity between trees, and minimize the number of trees by selecting an optimal subset of the forest. While these studies demonstrate that increasing diversity is crucial for improving random forest generalization, the majority of existing methods focus on increasing diversity during the data sampling or tree-building stages. In contrast, our work explores a novel avenue by investigating how tree diversity can be enhanced after the trees have been constructed, specifically through the aggregation stage.

4. Dataset and Features

For our performance analysis, we used “Student Performance & Academic Trends Dataset” containing 8,000 student instances with 25 attributes. This dataset integrates demographic information, detailed measurements of AI usage behavior, study habits, engagement patterns, and academic outcomes. The dataset was publicly available and obtained from Kaggle [5]. The class variable is *performance_category*, a categorical variable with values “low,” “medium,” and “high,” indicating a student’s overall performance level in the class. Prior to model training, several preprocessing steps were applied. The unique identifier *student_id* was removed, as it provides no predictive information. The attributes *final_score* and *passed* were removed, as they directly correlate with the class variable to indicate the performance categories. The dataset showed a clear class imbalance, with Medium being the majority class (4705 samples), followed by Low (2,542 samples), and High as a minority class (753 samples).

Ranked attributes:	
0.25235	13 last_exam_score
0.14556	14 assignment_scores_avg
0.12141	16 concept_understanding_score
0.03134	10 ai_generated_content_percentage
0.01656	21 tutoring_hours
0.01535	19 sleep_hours
0.01077	9 ai_dependency_score
0.01062	7 ai_tools_used
0.00956	17 study_consistency_index
0.00909	12 ai_ethics_score
0.00794	3 grade_level
0.00743	15 attendance_percentage
0.00727	11 ai_prompts_per_week
0.00636	4 study_hours_per_day
0.00623	8 ai_usage_purpose
0.00585	18 improvement_rate
0.0052	5 uses_ai
0.00433	1 age
0.00357	20 social_media_hours
0.00347	22 class_participation_score
0.00228	2 gender
0.00106	6 ai_usage_time_minutes

Figure 1. Correlation Attribute Eval Results

For attribute selection, the Correlation Attribute Eval with the Ranker search was used (Figure 1). After applying an arbitrary cutoff of 0.008, we removed the last 12 attributes, resulting in 10 remaining attributes. Finally, the dataset was divided into three subsets: 70% of the data (5,600 students) for training, 15% (1,200 students) for validation, and 15% (1,200 students) for testing.

5. Methods

Our approach uses a Random Forest classifier as a base algorithm and proposes two diversity-aware variants that differ in their aggregation strategy.

5.1 Decision Tree

A decision tree recursively splits the training data to make each subset more pure with respect to the class label. At each node, the algorithm selects a feature based on maximum information gain and performs a split that best reduces the class uncertainty according to an

impurity criterion, which in our case is entropy. Then, each leaf node predicts the class based on the training samples that fall into that leaf.

5.2 Random Forest baseline

A Random Forest is an ensemble of decision trees trained on randomly selected bootstrap samples of training instances. Random feature selection for each sample additionally reduces correlation among trees and improves generalization. In the baseline model, all trees contribute equally, as the predictions are aggregated using uniform majority voting.

5.3 Diversity-aware voting using prediction agreement

$$A_{ij} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[y_i(x_m) = y_j(x_m)]$$

Equation 1. Pairwise Similarity. M =number of samples;
 y_i =class prediction of trees i and j for sample m

$$D_i^{\text{agree}} = 1 - \frac{1}{N-1} \sum_{j \neq i} A_{ij}$$

Equation 2. Prediction Agreement Diversity Score.
 N =number of trees

Our first proposed method assigns a diversity score to each tree based on how differently it behaves compared to the rest of the ensemble. After training the Random Forest, we evaluate each tree’s predictions on the validation set. We compute pairwise similarity (A_{ij}) (Equation 1) between trees by measuring how often two trees predict the same class on the validation examples: 0 if the predictions differ and 1 if they agree. A tree will have a higher diversity score (D_i^{agree}) when it has lower average agreement with other trees, meaning that it contributes less redundant predictions. The diversity scores are then normalized into weights, and we replace uniform aggregation with weighted aggregation. Therefore, trees with higher diversity weights have greater influence on the final ensemble prediction.

5.4 Diversity-aware voting using structural similarity

$$\hat{f}_i = \frac{f_i}{\sum_{k=1}^d c_{ik}}$$

Equation 3. Normalized fingerprint
vector of tree i . c_{ik} =# times feature k is
used to split in tree i ; d =# features

$$\text{sim}_{ij} = \frac{\hat{f}_i \cdot \hat{f}_j}{\|\hat{f}_i\| \|\hat{f}_j\|}$$

Equation 4. Cosine similarity of
normalized fingerprint vectors.

$$D_i^{\text{struct}} = 1 - \frac{1}{N-1} \sum_{j \neq i} \cos(f_i, f_j)$$

Equation 5. Structural Diversity
Score. N =number of trees

The second proposed method measures diversity using tree structure rather than prediction behavior. For each decision tree, we build a structural “fingerprint” vector (f_i) that counts how frequently each feature is used in splits within that tree, and we normalize this (Equation 3) so that trees of different sizes can be compared. We then calculate cosine similarity (Equation 4) between trees. Trees that rely on different sets of features, or use features in different proportions, are more structurally diverse. Each tree’s diversity score (D_i^{struct}) is calculated (Equation 5), then converted into normalized weights used to aggregate predictions.

6. Experiments/Results/Discussion

6.1 Parameters

The Random Forest was trained using 100 trees to reduce variance and make the ensemble’s predictions more stable. The maximum tree depth was set to 10 so that the model could capture non-linear relationships among academic and AI-usage features, while still limiting complexity to prevent overfitting. Entropy was used as the splitting criterion to measure class uncertainty in the multi-class setting. To address the class imbalance, we set the parameter `class_weight="balanced"` in the Random Forest classifier. This option automatically assigns each class a weight inversely proportional to its frequency in the training data, so that errors on rare classes are penalized more heavily than errors on common classes.

6.2 Performance metrics

Accuracy measures the overall proportion of correctly classified instances out of all test samples. Precision measures how many of the instances predicted as a certain class are actually correct. Recall, or sensitivity, measures how many true instances of a class are correctly identified by the model, indicating how well the model avoids missing rare but important cases. F1-score is the harmonic mean of precision and recall that balances both types of error. To account for class imbalance, we report both the macro-averaged F1-score, which treats all classes equally, and the weighted F1-score, which weights each class by frequency. Balanced Accuracy is the average of recall across all classes, making it more suitable than standard accuracy for this imbalanced multi-class problem. Cohen’s Kappa measures agreement between predicted and true labels while accounting for chance, giving a more reliable assessment than accuracy alone. Log Loss (cross-entropy) evaluates how well the predicted probability distributions match the true labels, reflecting the model’s confidence and calibration.

6.3 Results and Discussion

Metric	Baseline RF	Agreement-Weighted RF	Structural-Weighted RF
Accuracy	0.7658	0.7658	0.7667
Balanced Accuracy	0.7355	0.7359	0.7364
Macro F1	0.7167	0.7164	0.7170
Weighted F1	0.7677	0.7678	0.7687
Macro Precision	0.7026	0.7018	0.7025
Macro Recall	0.7355	0.7359	0.7364
Cohen Kappa	0.5829	0.5834	0.5847
Log Loss	0.5192	0.5196	0.5204

Table 1. Performance metrics

Baseline				Agreement weighted				Structurally weighted			
pred→	High	Low	Med	pred→	High	Low	Med	pred→	High	Low	Med
High	71	0	42	High	71	0	42	High	71	0	42
Low	0	313	69	Low	0	314	68	Low	0	314	68
Med	61	109	535	Med	62	109	534	Med	62	108	535

Table 2. Confusion matrices

As shown in Table 1, the diversity aware aggregation methods show small but consistent improvements over the baseline Random Forest model. The structural-weighted model achieves the highest accuracy (0.7667), balanced accuracy (0.7364), macro F1-score (0.7170), weighted F1-score (0.7687), and Cohen’s Kappa (0.5847), overall outperforming both the baseline and agreement-weighted models.

In addition to single summary metrics, we analyzed how performance changes as a function of ensemble diversity and the weighting parameter α , which is the exponent applied to each tree’s diversity score when computing its voting weight. Larger values of α amplify the votes of the most diverse trees. By varying α and plotting balanced accuracy against both α and the resulting diversity, we observe systematic trends that are not visible from final metrics alone.

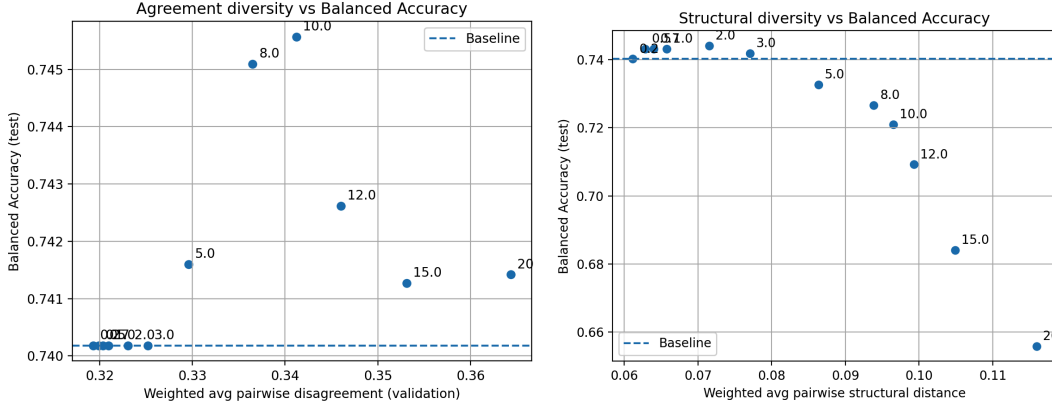


Figure 2. Effect of diversity on balanced accuracy

Figure 2 shows how increasing diversity weighting affects ensemble performance. The left plot shows the effect of agreement-based diversity on balanced accuracy. Moderate increases in disagreement ($\alpha \approx 8-10$) slightly improve balanced accuracy, showing that emphasizing trees that make different predictions better covers minority classes. The right plot shows the same analysis for structural diversity, showing that small increases in structural diversity ($\alpha \approx 0.5-3$) produce marginal gains, but overweighting considerably reduces performance.

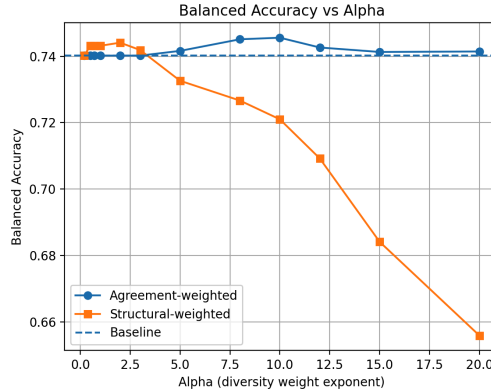


Figure 3. Balanced Accuracy vs. Alpha (diversity weight exponent)

Figure 3 directly plots balanced accuracy against α . Agreement-weighted voting shows a moderate peak around $\alpha \approx 8-10$, while structural weighting becomes unstable as α grows, falling well below the baseline. Together, these results show that mildly increasing diversity weight can slightly improve class-balanced performance, but excessive reweighting reduces generalization.

7. Conclusion/Future Work

In this work, we examined how diversity can be more effectively used in Random Forests by modifying the aggregation stage rather than the tree construction process. We introduced two diversity-aware ensemble variants that replace uniform majority voting with weighted voting based on prediction agreement and structural similarity among trees. Experiments on the Student Performance dataset demonstrated that both approaches consistently outperformed the baseline Random Forest across multiple evaluation metrics, including accuracy, balanced accuracy, F1-scores, and Cohen’s Kappa. Among the evaluated models, the structural-similarity Random Forest achieved the strongest overall performance, indicating that encouraging structural diversity among trees can reduce redundancy and improve generalization. For future research, we would explore the combination of multiple diversity metrics into a single weighted strategy and evaluate this approach on varied datasets. With additional computational resources, extending diversity aware aggregation would further improve the generality of random forests.

8. Contributions

In our project, we both worked on finding the dataset, writing the paper, and deciding our diversity metrics. Daksha worked on the data preprocessing and feature selection process. Ayla worked on training the models and generating graphs for the results.

9. Bibliography

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